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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/123,188	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/17/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-24 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-23 is/are rejected.

8) ☑ Claim(s) 24 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 03/17/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-24*** are pending.

2. ***Claims 1-23 are rejected.***

3. ***Claim 24 is objected.***

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

19. ***Claims 1 and 4-7 are rejected*** under 35 U.S.C. 102(a)(2) as being anticipated by Adams et al (Patent. No.: US 11,386,229 B2) hereinafter Adams.

As per claim 1, Adams discloses a vehicle (Adams, Fig 3, item 104)***, comprising:***

- a policy manager circuit (Adams, Fig 2-3, item 104, 106, 112) configured to interpret a data collection policy including a trigger condition (Adams, Fig 3, item 328) and a trigger evaluation data identifier (Adams, Fig 3, Col 3 lines: 15-27, Col 4 lines: 31-62, wherein the trigger evaluation data identifier can be one or more of the location identifier, the PIA policy identifier, the key identifier and/or the trigger identifier. The system 300 shows an example of an enforcement policy 321 defines the region 322 to which enforcement policy is applicable. The example provides defines the region based upon a country, however small subdivisions or geographic coordinates can be utilized. The enforcement policy 321 can define the types of objects 324 or data that must be filtered or redacted within the contents and the means by

which the data must be protected. For example, the entry '1' may define be associated with redacting only faces within an image. In addition, the encryption key 326, or a location of encryption key, may be provided within the enforcement policy. Triggers 328 may also be defined as to when data is to be filtered for example at the time of data collection, when the data is uploaded to the network, or when a trigger event associated with the vehicle occurs such as an accident, public alert or privacy identification sign is received requesting information from vehicles that are within a defined area);

- an endpoint configured to capture a trigger evaluation data stream in response to the trigger evaluation data identifier and the trigger condition (Adams, Fig 2-3, Col 4 lines: 31-62, wherein the endpoint can be the component within the vehicle that collects data in response to the trigger condition and based on one or more of the location identifier, the PIA policy identifier, the key identifier and/or the trigger identifier. The system 300 shows an example of an enforcement policy 321 defines the region 322 to which enforcement policy is applicable. The example provides defines the region based upon a country, however small subdivisions or geographic coordinates can be utilized. The enforcement policy 321 can define the types of objects 324 or data that must be filtered or redacted within the contents and the means by which the data must be protected. For example, the entry '1' may define be associated with redacting only faces within an image. In addition, the encryption key 326, or a location of encryption key, may be provided within the enforcement policy. Triggers 328 may also be defined as to when data is to be filtered for example at the time of data collection, when the data is uploaded to the network, or when a trigger event associated with the vehicle occurs such as an accident, public alert or privacy identification sign is received requesting information from vehicles that are within a defined area.); and

- a vehicle data interface configured to receive the trigger evaluation data stream

{Adams, Fig 2-3, Col 4 lines: 4-30, wherein the system 106 comprises one or more processors 202 that are coupled to a memory 204. The instructions stored in a storage component or device 206 are executed by the processor 202 from the memory 204. The instructions provide functionality for video/image/data processing 224 which provides for the filtering of PII objects 222 that are associated with an enforcement policy 224. The processor 202 is coupled to one or more sensors 210a-210f of the vehicle 104 through a sensor subsystem 210. These sensors may include, but not be limited to, LiDAR 210a, radar 210b, one or more cameras 210c, gyroscopes 210d, accelerometers 210e or ultrasonic devices 210f. The data from these devices is collected by the processor 202 and processed using the functionality 220. The enforcement policy 224 is received through a communication subsystem 212 that may contain a system identity module (SIM) 214 via one or more antennas 216. The vehicle may utilize cellular or mobile networks to transmit and receive data. Alternatively, local area networks or short-range networks may be utilized to upload or download the data as required};

As pre claim 4, claim 1 is incorporated and Adams discloses **wherein the data collection policy includes a plurality of trigger evaluation data identifiers configured to identify trigger evaluation to be evaluated by a plurality of trigger conditions of the data collection policy, wherein the plurality of trigger evaluation data identifiers correspond to a plurality of data types including at least two of a controller area network (CAN) message, a CAN signal, an ethernet packet, a vehicle location, a vehicle status, and a diagnostic trouble code** {Adams, Fig 3, Col 3 lines: 15-27, Col 4 lines: 31-62, wherein the trigger evaluation data identifier can be one or more of the location identifier, the PIA policy identifier, the key identifier and/or the trigger identifier. As shown in Fig 3, there are multiple trigger evaluation data identifiers corresponding to at least a vehicle location and a vehicle status (for example the trigger indicating that the car

had an accident). The system 300 shows an example of an enforcement policy 321 defines the region 322 to which enforcement policy is applicable. The example provides defines the region based upon a country, however small subdivisions or geographic coordinates can be utilized. The enforcement policy 321 can define the types of objects 324 or data that must be filtered or redacted within the contents and the means by which the data must be protected. For example, the entry '1' may define be associated with redacting only faces within an image. In addition, the encryption key 326, or a location of encryption key, may be provided within the enforcement policy. Triggers 328 may also be defined as to when data is to be filtered for example at the time of data collection, when the data is uploaded to the network, or when a trigger event associated with the vehicle occurs such as an accident, public alert or privacy identification sign is received requesting information from vehicles that are within a defined area).

As pre claim 5, claim 1 is incorporated and Adams discloses wherein the data collection policy includes a plurality of trigger evaluation data identifiers configured to identify trigger evaluation to be evaluated by a plurality of trigger conditions of the data collection policy, wherein the plurality of trigger evaluation data identifiers correspond to a plurality of data types including at least one of a controller area network (CAN) message, a CAN signal, an ethernet packet, a vehicle location, a vehicle status, and a diagnostic trouble code (Adams, Fig 3, Col 3 lines: 15-27, Col 4 lines: 31-62, wherein the trigger evaluation data identifier can be one or more of the location identifier, the PIA policy identifier, the key identifier and/or the trigger identifier. As shown in Fig 3, there are multiple trigger evaluation data identifiers corresponding to at least a vehicle location and a vehicle status (for example the trigger indicating that the car had an accident). The system 300 shows an example of an enforcement policy 321 defines the

region 322 to which enforcement policy is applicable. The example provides defines the region based upon a country, however small subdivisions or geographic coordinates can be utilized. The enforcement policy 321 can define the types of objects 324 or data that must be filtered or redacted within the contents and the means by which the data must be protected. For example, the entry '1' may define be associated with redacting only faces within an image. In addition, the encryption key 326, or a location of encryption key, may be provided within the enforcement policy. Triggers 328 may also be defined as to when data is to be filtered for example at the time of data collection, when the data is uploaded to the network, or when a trigger event associated with the vehicle occurs such as an accident, public alert or privacy identification sign is received requesting information from vehicles that are within a defined area).

As pre claim 6, claim 1 is incorporated and Adams discloses wherein capturing the trigger evaluation data stream includes at least one of: receiving the trigger evaluation data stream from a data source communicatively coupled to the endpoint, or requesting the trigger evaluation data stream from a data source communicatively coupled to the endpoint (Adams, Fig 3, Col 3 lines: 15-27, Col 4 lines: 31-62, wherein the triggers 328 may also be defined as to when data is to be filtered for example at the time of data collection, when the data is uploaded to the network, or when a trigger event associated with the vehicle occurs such as an accident, public alert or privacy identification sign is received requesting information from vehicles that are within a defined area).

As pre claim 7, claim 1 is incorporated and Adams discloses wherein at least a portion of the captured trigger evaluation data stream is generated by the vehicle before the trigger

condition occurs. (Adams, Fig 3, Col 5 lines: 10-18, wherein based upon the determined region an associated enforcement policy can be retrieved either locally from a memory on the device or remotely from a server coupled by the wireless network (404). PII present with in the data can then be filtered based upon the associated enforcement policy requirements for example objects such as license plates, faces and location data may be removed (406). Thus, at least part of the vehicle data and/or the enforcement policy is captured before the trigger condition occurred).

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

20. ***Claims 10, 11-12 and 14-23 are rejected*** under 35 U.S.C. 102(a)(2) as being anticipated by Misu et al (Patent. No.: US 10,759,424 B2) hereinafter Misu.

As per claim 10, Misu discloses a vehicle (Misu, Fig 1, item 102, Fig 3)***, comprising:***

- a policy manager circuit (Misu, Fig 3) configured to interpret a data collection policy including a vehicle data identifier and a trigger configured to define a trigger condition (Misu, Fig 3, 15-16, Col 25 lines: 6-14, wherein capturing data within a window 1404 when a physiological event is detected in accordance with one aspect of the present disclosure. Fewer or more processes may be provided within the flow chart, and those shown should not be construed as limiting. The processes may begin at block 1500. A number of variants for

capturing data to be sent to the remote system);

- a trigger evaluation circuit (Misu, Fig 3) configured to:

- determine a trigger event occurrence in response to a trigger condition and trigger evaluation data (Misu, Fig 15, Col 25 lines: 15-28, wherein a physiological event determination is made. For example, and as shown above, the heart rate of the driver may be determined and if variations exist showing a stressed heart rate, a physiological event may be triggered. Increases or decreases in the driver's temperature may also indicate an event. Inflections in voice patterns may cause the event to occur. Facial monitoring may be used to also determine an event);

- determine a trigger event termination (Misu, Fig 15, Col 25 lines: 42-49, wherein to create data window 1404, at block 1510, a time frame is taken which includes a plus or minus time from the time of event. This vehicle data may show information which may be pertinent before the event was detected. As described above, the plus or minus time period does not have to be the same and may vary depending on the physiological event detected, how long the event lasts),

- determine a data capture window in response to the trigger event occurrence, trigger event termination, and the data collection policy (Misu, Fig 15, Col 25 lines: 42-59, wherein to create data window 1404, at block 1510, a time frame is taken which includes a plus or minus time from the time of event. This vehicle data may show information which may be pertinent before the event was detected. As described above, the plus or minus time period does not have to be the same and may vary depending on the physiological event detected, how long the event lasts, the repetitiveness of the event, etc. the vehicle data is captured in a window 1404. The data may be further refined or processed, for example, depending on the triggering event, the data may be refined to include data related to the event. Vehicle data related to some

control systems 346 may not be relevant when the event indicates that only vehicle data from gathering systems 336 are used. In one example, an event caused by a heart rate increase may not require vehicle data from the audio system 348 to modify automated driving functionalities), and

- capture identified vehicle data in response to the data capture window and the vehicle data identifier (Misu, Fig 15, Col 25 lines: 42-59, wherein to create data window 1404, at block 1510, a time frame is taken which includes a plus or minus time from the time of event. This vehicle data may show information which may be pertinent before the event was detected. As described above, the plus or minus time period does not have to be the same and may vary depending on the physiological event detected, how long the event lasts, the repetitiveness of the event, etc. the vehicle data is captured in a window 1404. The data may be further refined or processed, for example, depending on the triggering event, the data may be refined to include data related to the event. Vehicle data related to some control systems 346 may not be relevant when the event indicates that only vehicle data from gathering systems 336 are used. In one example, an event caused by a heart rate increase may not require vehicle data from the audio system 348 to modify automated driving functionalities);

As pre claim 11, claim 10 is incorporated and Misu discloses wherein the trigger evaluation circuit determines the trigger event occurrence by evaluating the trigger evaluation data using the trigger condition, wherein the trigger condition defines a relationship with a present value that may be satisfied or unsatisfied by the trigger evaluation data (Misu, Fig 3, Col 5 lines: 60-65, wherein for example, a physiological event of a driver's temperature reaching below a threshold may cause the vehicle to send vehicle data in the form of takeover data to the

remote system. The decrease of temperature may be used to determine whether a driver is sleeping as a driver's body temperature decreases in such a state).

As pre claim 12, claim 10 is incorporated and Misu discloses wherein the trigger evaluation data includes at least one of a vehicle state, a vehicle status, a vehicle operating mode, or a vehicle discrete event (Misu, Fig 3, Col 5 lines: 60-65, wherein for example, a physiological event of a driver's temperature reaching below a threshold may cause the vehicle to send vehicle data in the form of takeover data to the remote system. The decrease of temperature may be used to determine whether a driver is sleeping as a driver's body temperature decreases in such a state).

As pre claim 14, claim 10 is incorporated and Misu discloses wherein the data collection policy includes a plurality of trigger conditions, each of the plurality of trigger conditions corresponding to one of a plurality of trigger evaluation data values (Misu, Fig 3, Col 10 lines: 59-67, wherein Multiple applications 328 may run processes for obtaining vehicle data as well as detecting physiological events. For example, an application 328 within the operations 322 may be used solely to determine whether the driver is happy or nervous and another application 328 may be used for pulling steering information. Applications 328 may interact with one another to determine whether a physiological event has occurred. For example, seat information may be combined with facial expressions to conclude that a driver is hot).

As pre claim 15, claim 14 is incorporated and Misu discloses wherein the trigger evaluation circuit determines the trigger event occurrence in response to at least two of the plurality of trigger conditions (Misu, Fig 3, Col 10 lines: 59-67, wherein Multiple applications

328 may run processes for obtaining vehicle data as well as detecting physiological events. For example, an application 328 within the operations 322 may be used solely to determine whether the driver is happy or nervous and another application 328 may be used for pulling steering information. Applications 328 may interact with one another to determine whether a physiological event has occurred. For example, seat information may be combined with facial expressions to conclude that a driver is hot).

As pre claim 16, claim 14 is incorporated and Misu discloses **wherein: the data collection policy includes a trigger condition that corresponds to one of a plurality of trigger evaluation data values; and the trigger evaluation circuit determines the trigger event occurrence in response to the trigger condition** (Misu, Fig 3, Col 10 lines: 59-67, wherein Multiple applications 328 may run processes for obtaining vehicle data as well as detecting physiological events. For example, an application 328 within the operations 322 may be used solely to determine whether the driver is happy or nervous and another application 328 may be used for pulling steering information. Applications 328 may interact with one another to determine whether a physiological event has occurred. For example, seat information may be combined with facial expressions to conclude that a driver is hot).

As pre claim 17, claim 14 is incorporated and Misu discloses **wherein the plurality of trigger evaluation data values correspond to a plurality of data types, including at least two of a controller area network (CAN) message, a CAN signal, an ethernet packet, a vehicle location, a vehicle status, and a diagnostic trouble code** (Misu, Fig 3, Col 10 lines: 59-67, wherein Multiple applications 328 may run processes for obtaining vehicle data as well as detecting physiological events. For example, an application 328 within the operations 322 may be used

solely to determine whether the driver is happy or nervous and another application 328 may be used for pulling steering information. Applications 328 may interact with one another to determine whether a physiological event has occurred. For example, seat information may be combined with facial expressions to conclude that a driver is hot).

As pre claim 18, claim 14 is incorporated and Misu discloses wherein the plurality of trigger evaluation data values correspond to a plurality of data types, including at least one of a controller area network (CAN) message, a CAN signal, an ethernet packet, a vehicle location, a vehicle status, and a diagnostic trouble code (Misu, Fig 3, Col 10 lines: 59-67, wherein Multiple applications 328 may run processes for obtaining vehicle data as well as detecting physiological events. For example, an application 328 within the operations 322 may be used solely to determine whether the driver is happy or nervous and another application 328 may be used for pulling steering information. Applications 328 may interact with one another to determine whether a physiological event has occurred. For example, seat information may be combined with facial expressions to conclude that a driver is hot).

As pre claim 19, claim 10 is incorporated and Misu discloses wherein the trigger evaluation data includes a virtual sensor value derived from a plurality of vehicle data values (Misu, Fig 3, Col 10 lines: 59-67, wherein Multiple applications 328 may run processes for obtaining vehicle data as well as detecting physiological events. For example, an application 328 within the operations 322 may be used solely to determine whether the driver is happy or nervous and another application 328 may be used for pulling steering information. Applications 328 may interact with one another to determine whether a physiological event has occurred.

For example, seat information may be combined with facial expressions to conclude that a driver is hot).

As pre claim 20, claim 10 is incorporated and Misu discloses **wherein the data capture window terminates after a delay following the trigger event termination in response to the data collection policy** (Misu, Fig 3, Col 23 lines: 30-34, wherein an appropriate amount of time may elapse before sending the vehicle data to the remote system 104. The time period may be used to remove any inaccurate readings through the heart monitors).

As pre claim 21, claim 10 is incorporated and Misu discloses **wherein the data capture window terminates based at least in part on one or more trigger conditions** (Misu, Fig 3, Col 23 lines: 30-40, wherein an appropriate amount of time may elapse before sending the vehicle data to the remote system 104. The time period may be used to remove any inaccurate readings through the heart monitors. Alternatively, the vehicle data may be sent regardless of time period such that when it is detected the vehicle data is automatically sent. Different measurements in the heart rate may be related to different physiological states, for example, the driver may be upset, tired, lonely, etc. Each of these states may be used for providing different types of vehicle data to the remote system 104).

As pre claim 22, claim 10 is incorporated and Misu discloses **wherein the data collection policy includes a plurality of trigger conditions having a plurality of trigger types, the plurality of trigger types including at least two of a signal trigger, a vehicle status trigger, a timing trigger, a schedule trigger, a geofence trigger, an error trigger, an environment trigger, or a user input trigger** (Misu, Fig 3, Col 10 lines: 59-67, wherein Multiple applications 328 may run

processes for obtaining vehicle data as well as detecting physiological events. For example, an application 328 within the operations 322 may be used solely to determine whether the driver is happy or nervous and another application 328 may be used for pulling steering information. Applications 328 may interact with one another to determine whether a physiological event has occurred. For example, seat information may be combined with facial expressions to conclude that a driver is hot).

As pre claim 23, claim 22 is incorporated and Misu discloses wherein determining the trigger event occurrence is based on a first portion of the plurality of trigger conditions, the first portion including a first trigger having a first trigger type and a second trigger having a second trigger type (Misu, Fig 3, Col 10 lines: 59-67, wherein Multiple applications 328 may run processes for obtaining vehicle data as well as detecting physiological events. For example, an application 328 within the operations 322 may be used solely to determine whether the driver is happy or nervous and another application 328 may be used for pulling steering information. Applications 328 may interact with one another to determine whether a physiological event has occurred. For example, seat information may be combined with facial expressions to conclude that a driver is hot).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention

and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

21. ***Claim***(s) **2-3** is/are rejected under 35 U.S.C. 103 as being unpatentable over Adams et al (Patent. No.: US 11,386,229 B2) hereinafter Adams in view of Addepalli et al (Pub. No.: US 2014/0215491 A1).

As pre claim 2, claim 1 is incorporated and Adams discloses **a plurality of endpoints including a first endpoint communicatively, and wherein at least a portion of the plurality of endpoints are configured to capture a plurality of trigger evaluation data values of the trigger evaluation data stream in response to the data collection policy** (Adams, Fig 2-3, Col 4 lines: 4-30, wherein the plurality of endpoints can be the plurality of sensors within the vehicle. The system 106 comprises one or more processors 202 that are coupled to a memory 204. The instructions stored in a storage component or device 206 are executed by the processor 202 from the memory 204. The instructions provide functionality for video/image/data processing 224 which provides for the filtering of PII objects 222 that are associated with an enforcement policy 224. The processor 202 is coupled to one or more sensors 210a-210f of the vehicle 104 through a sensor subsystem 210. These sensors may include, but not be limited to, LiDAR 210a, radar 210b, one or more cameras 210c, gyroscopes 210d, accelerometers 210e or ultrasonic devices 210f. The data from these devices is collected by the processor 202 and processed using the functionality 220. The enforcement policy 224 is received through a communication subsystem 212 that may contain a system identity module (SIM) 214 via one or more antennas 216. The vehicle may utilize cellular or mobile networks to transmit and receive data. Alternatively, local area networks or short-range networks may be utilized to upload or download the data as required). ***Adams does not explicitly disclose that the plurality of endpoints are coupled to an ethernet switch.*** However, using an ethernet switch to couple

multiple endpoints is well known in the art. For example, Addepalli discloses ***that the plurality of endpoints are*** **coupled to an ethernet switch** (Addepalli, Fig 1, paragraph 0038, wherein system in one example embodiment includes an internal subsystem in a vehicle including a plurality of sensors, actuators and vehicle controllers and an on-board unit (OBU) in communication with the internal subsystem on a first network, and a node on a second network. The first network includes at least one Ethernet network including a first Ethernet switch coupled to the OBU and adapted to facilitate communication between the plurality of sensors, actuators and vehicle controllers in the internal subsystem, and the second network comprises a wireless network. More specific embodiments include an electronic control unit (ECU) with at least one processing element and at least one processing logic and other features).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Adams in view of Addepalli such that an ethernet switch is used as claimed because this would have provided a way to provide the ability to conduct transactions in vehicular network environments in an optimized manner and providing a secure and flexible communication framework for various agents conducting the transactions (see Addepalli, paragraph 0003).

As pre claim 3, claim 2 is incorporated and Adams discloses **wherein the first endpoint is communicatively coupled to a plurality of networks configured to communicate using a plurality of communication protocols, and wherein the first endpoint is configured to capture a plurality of trigger evaluation data values of the trigger evaluation data stream from the plurality of networks in response to the data collection policy** (Adams, Fig 2-3, Col 4 lines: 4-30, wherein the plurality of endpoints can be the plurality of sensors within the vehicle. The system 106 comprises one or more processors 202 that are coupled to a memory 204. The instructions

stored in a storage component or device 206 are executed by the processor 202 from the memory 204. The instructions provide functionality for video/image/data processing 224 which provides for the filtering of PII objects 222 that are associated with an enforcement policy 224. The processor 202 is coupled to one or more sensors 210a-210f of the vehicle 104 through a sensor subsystem 210. These sensors may include, but not be limited to, LiDAR 210a, radar 210b, one or more cameras 210c, gyroscopes 210d, accelerometers 210e or ultrasonic devices 210f. The data from these devices is collected by the processor 202 and processed using the functionality 220. The enforcement policy 224 is received through a communication subsystem 212 that may contain a system identity module (SIM) 214 via one or more antennas 216. The vehicle may utilize cellular or mobile networks to transmit and receive data. Alternatively, local area networks or short-range networks may be utilized to upload or download the data as required).

22. ***Claim***(s) **8-9** is/are rejected under 35 U.S.C. 103 as being unpatentable over Adams et al (Patent. No.: US 11,386,229 B2) hereinafter Adams in view of Gaur et al (Pub. No.: US 2018/0232235 A1).

As pre claim 8, claim 7 is incorporated and ***Adams does not explicitly disclose a rotating buffer circuit structured to store the at least a portion of the captured trigger evaluation data stream.*** However, using a rotating buffer to store data is well known in the art. For example, Gaur discloses ***a rotating buffer circuit structured to store the at least a portion of the captured trigger evaluation data stream*** (Gaur, paragraph 0030, wherein the buffer 220 may be a first-in-first-out (FIFO) buffer, a rotating buffer, a ring buffer, or other type of channel-based storage).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Adams in view of Gaur such that a rotating buffer is used as claimed because this would have provided one well-known available method for storing data with limited memory size in an efficient matter.

As pre claim 9, claim 8 is incorporated and Adams discloses **wherein the rotating buffer circuit is further structured to store the at least a portion of the captured trigger evaluation data stream in a rotating time window** (Gaur, paragraph 0022, 0030, wherein determination may occur every "n" micro-operations within the DDG, and thus be performed for a rotating window of micro-operations passing through the buffer. The buffer 220 may be a first-in-first-out (FIFO) buffer, a rotating buffer, a ring buffer, or other type of channel-based storage).

23. ***Claim***(s) **13** is/are rejected under 35 U.S.C. 103 as being unpatentable over Misu et al (Patent. No.: US 10,759,424 B2) hereinafter Misu in view of Gaur et al (Pub. No.: US 2018/0232235 A1).

As pre claim 13, claim 10 is incorporated and Misu discloses **wherein the identified vehicle data is generated by the vehicle before the trigger event occurrence** (Misu, Fig 15, Col 25 lines: 42-59, wherein to create data window 1404, at block 1510, a time frame is taken which includes a plus or minus time from the time of event. This vehicle data may show information which may be pertinent before the event was detected. As described above, the plus or minus time period does not have to be the same and may vary depending on the physiological event detected, how long the event lasts, the repetitiveness of the event, etc. the vehicle data is captured in a window 1404) **stored using a buffer circuit** (Misu, Col 8 lines: 65-67, wherein the in-vehicle computing system 320 may buffer data to anticipate intermittent loss of connectivity

with the communication network 108 (e.g., when moving through an area with poor network coverage) or to save on network bandwidth costs). ***Misu does not explicitly disclose a rotating buffer circuit***. However, using a rotating buffer to store data is well known in the art. For example, Gaur discloses ***a rotating buffer circuit*** (Gaur, paragraph 0030, wherein the buffer 220 may be a first-in-first-out (FIFO) buffer, a rotating buffer, a ring buffer, or other type of channel-based storage).

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify Adams in view of Gaur such that a rotating buffer is used as claimed because this would have provided one well-known available method for storing data with limited memory size in an efficient matter.

Allowable Subject Matter

Claim 24 is objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached 7:30 am - 3:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/HAMZA N ALGIBHAH/
Primary Examiner, Art Unit 2441